

OverTheWire Bandit Level 0 to 15 Write-Up

Level 0 → 1 — Reading a File

Problem

After logging in as `bandit0`, I had to find the password for the next level. The password was stored in a file named `readme`.

Solution

I first used `ls` to check the files in the current directory and found the `readme` file.

Then I used `cat readme` to read the file, which gave me the password for the next level.

Screenshot

```
bandit0@bandit:~$ ls
readme
bandit0@bandit:~$ cat readme
Congratulations on your first steps into the bandit game!!
Please make sure you have read the rules at https://overthewire.org/rules/
If you are following a course, workshop, walkthrough or other educational activity,
please inform the instructor about the rules as well and encourage them to
contribute to the OverTheWire community so we can keep these games free!

The password you are looking for is: 6y2kwnwK6grgvwvpvLaa2T1cpFEK0hNR
bandit0@bandit:~$
```

Bandit Level 1 → 2 — Special Filename

Problem

The password was stored in a file named `-`. When I tried to open it normally, Linux treated `-` as standard input instead of a filename.

Solution

I used `./-` to tell Linux that I wanted the file named `-` from the current directory.

```
cat ./-
```

This displayed the password for the next level.

Screenshot

```
bandit1@bandit:~$ ls -la
total 24
-rw-r----- 1 bandit2 bandit1  33 Jun 24 14:59 -
drwxr-xr-x  2 root    root    4096 Jun 24 14:59 .
drwxr-xr-x 150 root    root    4096 Jun 24 15:02 ..
-rw-r--r--  1 root    root    220 Feb 13  2026 .bash_logout
-rw-r--r--  1 root    root   3851 Jun 24 14:50 .bashrc
-rw-r--r--  1 root    root    807 Feb 13  2026 .profile
bandit1@bandit:~$ cat ./-
PK8fYLZg2hnHSz83plBL1iEPKdD3QToB
bandit1@bandit:~$
```

Bandit Level 2 → 3 — Filename with Spaces

Problem

The password was inside a file whose name contained spaces. Typing the filename normally made Linux treat each word separately.

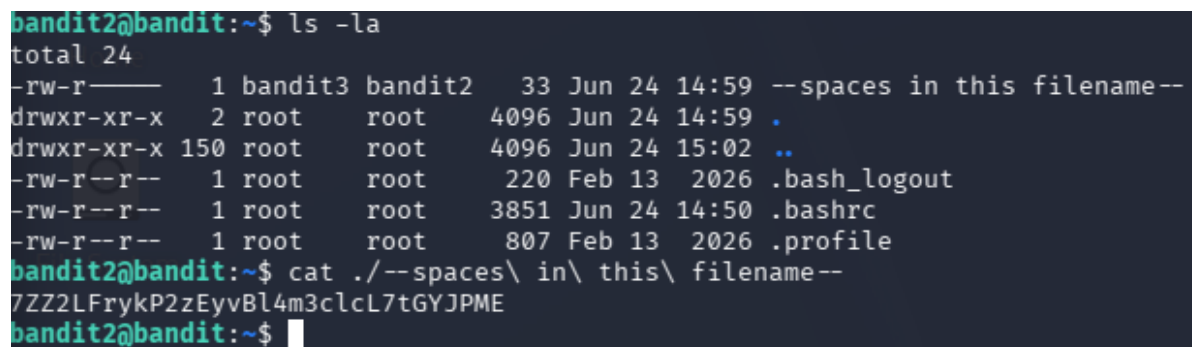
Solution

I placed the complete filename inside quotes.

```
cat ./--spaces\ in\ this\ filename--
```

This allowed me to read the file normally.

Screenshot



```
bandit2@bandit:~$ ls -la
total 24
-rw-r----- 1 bandit3 bandit2  33 Jun 24 14:59 --spaces in this filename--
drwxr-xr-x  2 root    root    4096 Jun 24 14:59 .
drwxr-xr-x 150 root    root    4096 Jun 24 15:02 ..
-rw-r--r--  1 root    root    220 Feb 13  2026 .bash_logout
-rw-r--r--  1 root    root   3851 Jun 24 14:50 .bashrc
-rw-r--r--  1 root    root    807 Feb 13  2026 .profile
bandit2@bandit:~$ cat ./--spaces\ in\ this\ filename--
7ZZ2LFrykP2zEyyB14m3clcL7tGYJPME
bandit2@bandit:~$
```

Bandit Level 3 → 4 — Hidden File

Problem

The password was stored in a hidden file inside the `inhere` directory. Normal `ls` did not show the hidden file.

Solution

I entered the directory and used `ls -la` to show all files, including hidden ones.

After finding the hidden file, I used `cat` to read it and got the password.

Screenshot

```
bandit3@bandit:~$ cd inhere
ls -la
total 12
drwxr-xr-x 2 root    root    4096 Jun 24 14:59 .
drwxr-xr-x 3 root    root    4096 Jun 24 14:59 ..
-rw-r----- 1 bandit4 bandit3   33 Jun 24 14:59 ... Hiding-From-You
bandit3@bandit:~/inhere$ cat ... Hiding-From-You
xZTXq1rDJQVVAzdV5cHq1TQytTWufAMq
bandit3@bandit:~/inhere$
```

Bandit Level 4 → 5 — Human-Readable File

Problem

There were multiple files inside the directory, but only one of them contained human-readable data.

Solution

Instead of opening every file manually, I used:

```
file ./*
```

This showed the type of each file. I found the file marked as ASCII text and used `cat` to read it.

Screenshot

```
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ file ./*
./-file00: data
./-file01: data
./-file02: data
./-file03: data
./-file04: data
./-file05: data
./-file06: OpenPGP Public Key
./-file07: ASCII text
./-file08: data
./-file09: Motorola S-Record; binary data in text format
bandit4@bandit:~/inhere$ cat ./-file07
6C7h9GD8M6ai5nr7wo1RonrzFjj9yIrG
bandit4@bandit:~/inhere$
```

Bandit Level 5 → 6 — Finding a File by Size

Problem

There were many files and directories. The required file was human-readable, exactly 1033 bytes, and not executable.

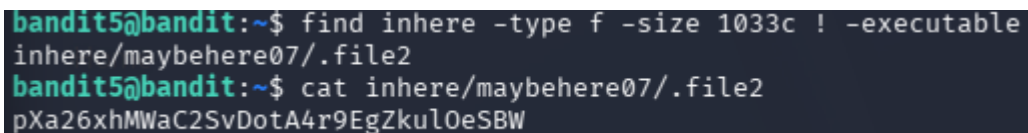
Solution

I used `find` with the properties given in the question.

```
find . -type f -size 1033c ! -executable
```

It returned the matching file, and I used `cat` to read it.

Screenshot



```
bandit5@bandit:~$ find inhere -type f -size 1033c ! -executable
inhere/maybeh ere07/.file2
bandit5@bandit:~$ cat inhere/maybeh ere07/.file2
pXa26xhMWaC2SvDotA4r9EgZkul0eSBW
```

Bandit Level 6 → 7 — Finding a File in the System

Problem

The password file was somewhere on the server. The challenge gave its owner, group and file size.

Solution

I used `find` starting from `/` and searched using the given properties.

The command initially showed many permission errors, so I used `2>/dev/null` to hide those errors.

After finding the correct file, I opened it using `cat`.

Screenshot

```
bandit6@bandit:~$ find / -type f -user bandit7 -group bandit6 -size 33c 2>/dev/null
/var/lib/dpkg/info/bandit7.password
bandit6@bandit:~$ cat /var/lib/dpkg/info/bandit7.password
Bmnnvf82KzQlfxgAI2dlzYbr1u9pr3E3
bandit6@bandit:~$
```

Bandit Level 7 → 8 — Searching Inside a File

Problem

The password was inside `data.txt` next to the word `millionth`. The file contained many lines, so checking it manually would take time.

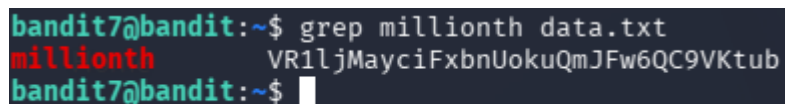
Solution

I used `grep` to directly search for the required word.

```
grep "millionth" data.txt
```

It showed the matching line along with the password.

Screenshot



```
bandit7@bandit:~$ grep millionth data.txt
millionth      VR1ljMayciFxbnUokuQmJFw6QC9VKtub
bandit7@bandit:~$
```


Bandit Level 8 → 9 — Finding the Unique Line

Problem

The password was the only line in `data.txt` that occurred once, while all the other lines were repeated.

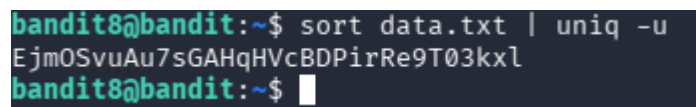
Solution

I first sorted the file so that duplicate lines were placed together. Then I used `uniq -u` to display only the unique line.

```
sort data.txt | uniq -u
```

The output gave me the password.

Screenshot



```
bandit8@bandit:~$ sort data.txt | uniq -u
EjmOSvuAu7sGAHqHVcBDPirRe9T03kx1
bandit8@bandit:~$
```

Bandit Level 9 → 10 — Finding Readable Text

Problem

The file contained mostly unreadable data. The password was one of the few readable strings and had several `=` characters before it.

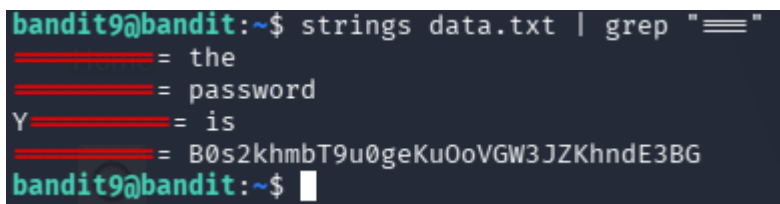
Solution

I used `strings` to extract the readable text and `grep` to filter the lines containing `=`.

```
strings data.txt | grep "=="
```

This helped me find the password without going through all the unreadable data.

Screenshot



```
bandit9@bandit:~$ strings data.txt | grep "=="
===== the
===== password
Y===== is
===== B0s2khmbT9u0geKuOoVGW3JZKhndE3BG
bandit9@bandit:~$
```

Bandit Level 10 → 11 — Base64

Problem

The contents of `data.txt` were encoded using Base64, so the actual password was not directly readable.

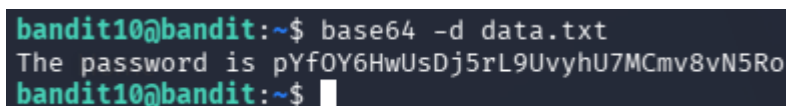
Solution

I decoded the file using:

```
base64 -d data.txt
```

The decoded output contained the password for the next level.

Screenshot



```
bandit10@bandit:~$ base64 -d data.txt
The password is pYf0Y6HwUsDj5rL9UvyhU7MCmv8vN5Ro
bandit10@bandit:~$
```

Bandit Level 11 → 12 — ROT13

Problem

The letters inside `data.txt` had been rotated by 13 positions using ROT13.

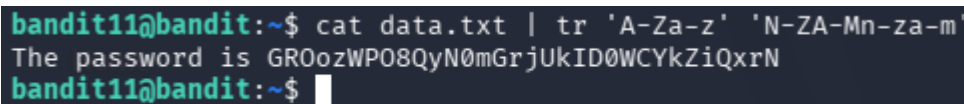
Solution

I used the `tr` command to convert the letters back to their original form.

```
tr 'A-Za-z' 'N-ZA-Mn-za-m' < data.txt
```

After decoding the text, I got the password.

Screenshot



```
bandit11@bandit:~$ cat data.txt | tr 'A-Za-z' 'N-ZA-Mn-za-m'
The password is GR00zWP08QyN0mGrjUkID0WCYkZiQxrN
bandit11@bandit:~$
```

Bandit Level 12 → 13 — Hexdump and Compression

Problem

This level was a little harder. `data.txt` was a hexdump of a file that had been compressed multiple times using different formats.

Solution

I first created a temporary directory and copied `data.txt` into it.

Then I reversed the hexdump using `xxd -r`.

After that, I repeatedly used `file` to check what type of compression I was dealing with.

Depending on the result, I used `gunzip`, `bunzip2`, or `tar` to extract the next layer.

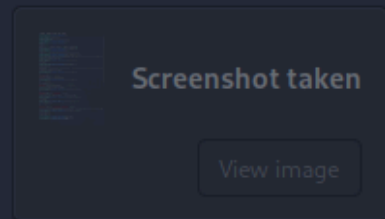
I kept repeating:

check file type → extract → check again

Eventually, I reached a normal text file containing the password.

Screenshot

```
bandit12@bandit:/tmp/tmp.A1w5nKFs8k$ cd
bandit12@bandit:~$ mkdir -p /tmp/tmp.FlBxP1uphS
bandit12@bandit:~$ cd /tmp/tmp.FlBxP1uphS
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ ls -ls
total 0
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ ls -la
total 0
drwx----- 2 bandit12 bandit12 40 Sep 10 12:21 .
drwxrwx-wt 115 root root 2960 Sep 10 12:21 ..
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ cp ~/data.txt .
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ ls -l
total 4
-rw-r----- 1 bandit12 bandit12 2641 Sep 10 12:22 data.txt
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ xxd -r data.txt > data
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ file data
data: gzip compressed data, was "data2.bin", last modified: Wed Jun 24 14:58:46 2026, ma
riginal size modulo 2^32 580
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ mv data data.gz
gunzip data.gz
file data
data: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ mv data data.bz2
bunzip2 data.bz2
file data
data: gzip compressed data, was "data4.bin", last modified: Wed Jun 24 14:58:46 2026, ma
riginal size modulo 2^32 20480
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ mv data data.tar
tar -xf data.tar
ls
file data5.bin
data.tar data.txt data5.bin
data5.bin: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ mv data data.tar
tar -xf data.tar
ls
file data5.bin
mv: cannot stat 'data': No such file or directory
data.tar data.txt data5.bin
data5.bin: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ mv data5.bin data5.tar
tar -xf data5.tar
ls
file data6.bin
data.tar data.txt data5.tar data6.bin
data6.bin: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ mv data6.bin data6.bz2
bunzip2 data6.bz2
file data6
data6: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ mv data6 data6.tar
tar -xf data6.tar
ls
file data8.bin
data.tar data.txt data5.tar data6.tar data8.bin
data8.bin: gzip compressed data, was "data9.bin", last modified: Wed Jun 24 14:58:46 202
ix, original size modulo 2^32 49
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ mv data8.bin data8.gz
gunzip data8.gz
file data8
data8: ASCII text
bandit12@bandit:/tmp/tmp.FlBxP1uphS$ cat data8
The password is qQYQiHOBPR8zR61qxYqX45quvihF2uzk
bandit12@bandit:/tmp/tmp.FlBxP1uphS$
```



Bandit Level 13 → 14 — SSH Private Key

Problem

Instead of giving the next password directly, this level provided an SSH private key named `sshkey.private`.

Solution

I found the private key and used it with the SSH `-i` option to authenticate as the next Bandit user.

The basic command was:

```
ssh -i sshkey.private bandit14@localhost -p 2220
```

I also learned that SSH private keys need to have proper file permissions and that the Bandit environment can have some connection restrictions.

Screenshot

```
bandit13@bandit:~$ ls -la
total 28
drwxr-xr-x  2 root    root    4096 Jun 24 14:58 .
drwxr-xr-x 150 root    root    4096 Jun 24 15:02 ..
-rw-r--r--  1 root    root     220 Feb 13  2026 .bash_logout
-rw-r--r--  1 root    root    3851 Jun 24 14:50 .bashrc
-rw-r--r--  1 root    root     807 Feb 13  2026 .profile
-rw-r----- 1 bandit14 bandit13 467 Jun 24 14:58 HINT
-rw-r----- 1 bandit14 bandit13 2602 Jun 24 14:58 sshkey.private
bandit13@bandit:~$ file sshkey.private
sshkey.private: OpenSSH private key
bandit13@bandit:~$ cat sshkey.private
-----BEGIN OPENSSH PRIVATE KEY-----
b3B1bnNzaC1rZXktdjEAAAABG5vbmUAAAABbm9uZQAAAAAAAAABAAAABlAAAAAdzc2gtcn
NhAAAAAwEAAQAAAYEAuCCoxSR6xDCnsm98Kqan1x1JF3mfZ6kZa+BoAHetQ/91F4EKTfME
E+Xs8yVgLn1YLOtVvyVswMzy330eFJV/KEzef54V8yZo7jFc+pqL0F6+BFRy+wsyLASV5
XxD/AtafdbVfLZFGTLC53k0FtT0VxokFmnTlIwRyxRJUNAWP9+fKAtbnfqkixWqg0ZaLr
1fICyam6vb61lnMfuIGHZyQNHETkOKZgAaMnQW6bYLRjnkxsNNxk1pj0sT3MUQdLrPCdyh
l8r4UqZ60NZPA6X82/hJNQJR+zVghXrGmTTLU7MAHN7rTdoq4bIQtdShuZmd5igp50vVg
tCn4bRM2CFX18m0P6gNDpR+VYH5f99MwdXyrDUAzuusH7H6CyawRZZF8rFog0Nc6HwTJ2h
YJwZuf85tD3wR8XmzvnB1f2+U90Emhs01uEYNTNvhq7Gvz/TGI0wR0hEBmsznSns0jShT
cqVsnTMhHaXse3u2D+tbGoC3w0C0yJeX7Skhq63bAAAFiMnVC+zJ1QvsAAAAB3NzaC1yc2
EAAAQBAJgggMukesQwp7JvfJKmp9cdSRd5n2epGWvgaAB3rUP/dREBck35h8Pl7PmLYC4Z
9WC9E78lBMDM8t9znhsVFyhm3n+eFfMma04xXMfQUJTHevgRUcvsLMiWELeV8Q/wLWn3W1
Xy2RRRkywud5DhX09FcaJBZp05SMEcsUSbjQFj/fn5ALW536pJIsVqoNGWi69XyAmGpur2+
opZjX7ohh2ckDR3k5DImYAGjJ0Fum2JUJ55MbDTcZNaY9LE9zFEHS6zwncoZfK11KmetDW
Tw0l/Nv4STUCUfS1YIV6xpk05VOzABze603aKuGyELQ0obmWZneYoKeTr1YLQp+G0TNghV
5fJd+oDQ6Uf1WB+X/fTMHV8qW1AM7rrB+x+gsmEWWrfKxaKjX0h8EydowCCNrn/OZbQ
98EFf1s7SwZ9v1PthJobNNbhGDbTb4auxr8/0xiNMEDIRAZrM50p7NI0oU3K1bJ0zIR2l
7Ht7tg/rWxqAt8DnDso3l+0pIaut2wAAAAMBAAEAAAGAEgTUL1LGfTwCFUK2Yk05gKIzjH
IRCPZx7+4qj10m31AqR84PgZj49W+LVDIkqu5MZpaqT4rsj50hYv+wCSimJH39SjTgxgZM
v361K0hBcYhtXchoHlIzAcLFUL/yMtKYxyV3UT5uQwIoIq9lbaQerP7jlrjHwDFP2y85k9
oqams6aEWEjKp8kKs/e/USB3c9qBbCZ+dRyI7W32vDKvZsB0puZC4JrYe0nqpmRY968LD6
3Lty3WtyDNQ0IgI/s/B1K4wmzFM/drZHOLJYjzJssyvjaSf5XydPq0Sk+z/0wRut1ev4+j
2hBuFqunZn0u2nIz4LSocLPtTZqu2PhbQgGJj80aXSH+TkKoLv+sG8d0+rrWtllkdJPX
YsWBmcWfd/Rj0LnkOBsGu+00ne4IpK+7wAWNoJwpw8MQkLRJkATay0rsGw3Bt7Bf7oqnDA
gPFNfgcKmfhWiORhwezgm5zEK50g4cCgwdCH2YQwQLuRQHKKPat9XjXcXs3j3N9D1BAAAA
wF2LTVT5VAvkqhJRe4KeT8RhydARmWdt/6YUY+liDnK/Bm4lhpeUrXJ2frXs1sGeAty60ZL
LA31Y+kV9xgS4HN0T2aRTCXdfVJ4XmMcsITxcRejTo92c3EIfd/ttbxkoLTsR/2tjkYje3
GbMOZpg3n0RgtPH0sjYfy5XyBBYNEg7GEJ6jSNBr9nA+wZILqVbuaDdn+FeVcgWSxFUYIC
kGZ0HketuIdv7EkerLJCckL0L16Z3i0jwLfcMuZj98b1u/QAAAMEA5JNGvvpj8WT4aQuM7
XME/N/m//8/sAb2DMHUMWY0ySSNRDbM3pJkKja7nUp0CybwTYTrf/agLxIgdQAO71GpK7
ePRmsGu5+mrrreNJKicGyER0IqCBhbWyljGDKFMJNGb2gk0Mu9XNZk9y3LRDmPRcfBMekmV
fAjvy5rGaqy/PsKznyZE12eGosH+CkfVWXDVRXd/ho0dWIZqaC+w8nLso6WLKhptWz8ke
SghZBBAsJW42cyenyvYdIzPvIcJpubAAAAwQ00Coj/QwJLwX0d75KuUsfgI0Esq9vLSWL
D53fQGhnNMEJZFD9B/W8Xa7VCLfPHDItcDoXiSgjdQnS1JeeVBBUHLULMLKMLNRNqjLD
0HSgyFYTpXl8t05ZAlFRNGEwe7y9RvNeXp9zPQC2v1FTVQ2RdALiWvR4fGyDu0ER4hY1A/
WfqaBSBABkdhylW1TBja/VncIHBZ04Bk1S9nwc0z6U5yTXpTwpZiJw1074Tjk+8X3or8WC
KN158egNsB+sEAAAA0cnVkeUBsb2NhbGhvc3QBAgMEBQ==
-----END OPENSSH PRIVATE KEY-----
bandit13@bandit:~$
```

Bandit Level 14 → 15 — Netcat

Problem

I had to submit the current password to a service running on `localhost` at port `30000`.

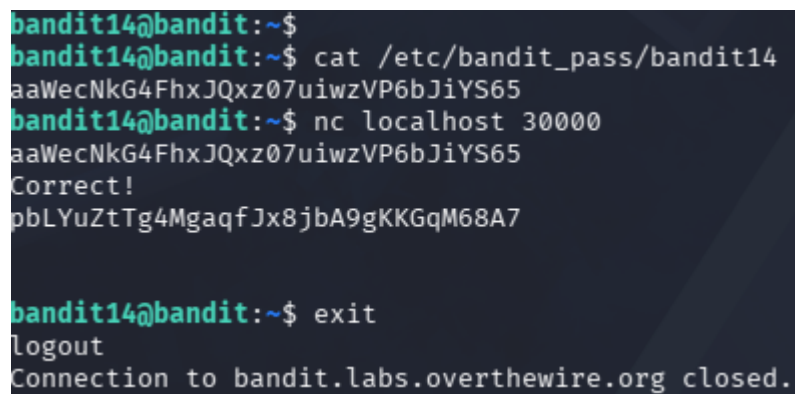
Solution

I connected to the port using Netcat.

```
nc localhost 30000
```

After entering the current password, the service returned the password for the next level.

Screenshot



```
bandit14@bandit:~$  
bandit14@bandit:~$ cat /etc/bandit_pass/bandit14  
aaWecNkG4FhxJQxz07uiwzVP6bJiYS65  
bandit14@bandit:~$ nc localhost 30000  
aaWecNkG4FhxJQxz07uiwzVP6bJiYS65  
Correct!  
pbLYuZtTg4MgaqfJx8jbA9gKKGqM68A7  
  
bandit14@bandit:~$ exit  
logout  
Connection to bandit.labs.overthewire.org closed.
```

Bandit Level 15 → 16 — SSL/TLS Connection

Problem

This level was similar to the previous one, but the service on port **30001** required an SSL/TLS encrypted connection.

Solution

Normal Netcat was not enough because the service expected TLS. So I used OpenSSL instead.

```
openssl s_client -connect localhost:30001
```

After connecting and entering the current password, I received the password for the next level.

Screenshot

```
bandit15@bandit:~$ cat /etc/bandit_pass/bandit15 | openssl s_client -connect localhost:30001 -quiet
Connecting to 127.0.0.1
Can't use SSL_get_servername
depth=0 CN=SnakeOil
verify error:num=18:self-signed certificate
verify return:1
depth=0 CN=SnakeOil
verify return:1
Correct!
kS0Hf0u5HiXFwKMKFqXvPd0TNGGa0X8V
bandit15@bandit:~$
```